

Probabilistic Dataset Merge

Scenario 5: MNN Not Confirmed

What happens when the matched row is even closer to a different target record

What this scenario demonstrates. In a real research run the target dataset contains many records, not just one. Mutual Nearest Neighbor (MNN) confirmation checks each match in both directions: the supplemental row matched to Target A is also compared against every other target, and if it sits even closer to some Target B, the pairing is one-sided. Note the algorithm has no way of knowing which pairing is *truly* correct — asymmetry is evidence for review, not proof of error. This scenario reproduces that condition with two nearly identical target records: the matched supplemental row sits between them on one feature, slightly closer to Target B, so the forward match to Target A is not confirmed in reverse.

The Target Record

This is the record we are trying to match. The system will search the supplemental dataset for the row closest to these values.

Record	Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units
Target A	2,469	40.2	99.5	649
Target B	2,467	40.6	99.5	649

Table 1: Target row values (raw, unstandardized). The match is performed for **Target A**.

Note: This scenario uses *two* target records to demonstrate MNN failure. Target A (shown above) has Avg. Wizard Age = 40.2; Target B (not shown) has Avg. Wizard Age = 40.6. The Rank 1 supplemental row has Avg. Wizard Age = 40.5, placing it marginally closer to Target B on that feature.

All Candidate Matches

The table below shows all 20 supplemental rows, sorted from closest to farthest. Distances are computed in standardized space (see Section 3). The **highlighted row** is the best match — the one the system selects.

Rank	Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units	Distance
1	2,468	40.5	99.5	649	0.0363
2	2,948	33.4	98.5	687	1.0324
3	2,524	30.0	99.5	557	1.2594
4	3,885	37.2	100.0	1,004	1.4623
5	3,849	42.2	99.5	1,043	1.4760
6	4,158	29.8	99.4	1,089	2.1101
7	4,555	26.5	99.4	930	2.3300
8	2,512	25.8	96.1	521	2.5068
9	4,822	25.7	98.2	914	2.5900
10	5,094	27.3	100.0	1,041	2.6411
11	4,891	34.4	99.6	1,339	2.6473
12	5,006	30.1	97.5	1,230	2.8763
13	5,296	25.4	99.3	1,128	2.9670
14	4,223	36.3	94.2	1,132	3.3335
15	4,701	58.7	100.0	1,519	3.6543
16	6,303	29.8	100.0	1,490	3.7643
17	3,465	47.9	92.5	1,080	4.0064
18	6,516	23.3	99.3	1,385	4.0256
19	6,668	32.0	98.5	1,641	4.1636
20	7,389	35.0	100.0	1,851	4.8246

Note: “Distance” is the standardized Euclidean distance between the target and each candidate row. A distance of 0.0000 means a perfect, identical match.

How Distance Is Calculated: A Worked Example

We use the **rank 2** candidate below to illustrate the calculation step by step. (The rank 1 match has distance 0 and is trivial; rank 2 shows the full arithmetic.)

Step 1 — Put every feature on the same scale (standardization).

Raw values are measured in different units (counts, years, percentages). Before computing distance we convert each to a *standardized score*: how many standard deviations above or below the combined average that value sits.

$$z = \frac{\text{raw value} - \text{combined average}}{\text{combined spread}}$$

Feature	Target (raw)	Candidate (raw)	Combined Avg.	Combined Spread
Dragon Sightings	2,469	2,948	4373.14	1450.23
Avg. Wizard Age	40.2	33.4	34.19	8.28
Pct. Leprechaun Cottages	99.5	98.5	98.64	1.92
Goblin Family Units	649	687	1069.45	357.50

Applying the formula above to each feature gives these standardized scores:

Feature	Target (std.)	Candidate (std.)
Dragon Sightings	-1.3130	-0.9827
Avg. Wizard Age	+0.7265	-0.0950
Pct. Leprechaun Cottages	+0.4493	-0.0709
Goblin Family Units	-1.1761	-1.0698

Step 2 — Compute the squared difference for each feature, then sum and take the square root.

$$d = \sqrt{\sum_i (z_{\text{target},i} - z_{\text{cand},i})^2}$$

Feature	Target (std.)	Cand. (std.)	Squared difference
Dragon Sightings	-1.3130	-0.9827	0.1091
Avg. Wizard Age	+0.7265	-0.0950	0.6749
Pct. Leprechaun Cottages	+0.4493	-0.0709	0.2706
Goblin Family Units	-1.1761	-1.0698	0.0113
<i>Sum</i>			<i>1.0659</i>

$$d = \sqrt{1.0659} = \mathbf{1.0324}$$

Distance Distribution

The chart below shows the distance from the target to each of the 20 candidate rows, sorted closest to farthest. The **highlighted bar** is the selected match.

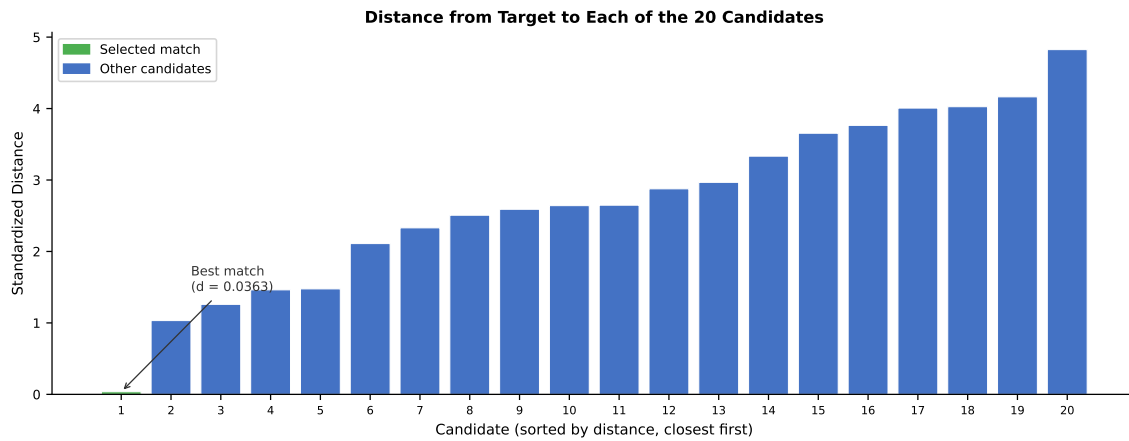


Figure 1: Distances from target to all 20 supplemental candidates, sorted ascending. The selected match is highlighted.

Signal Analysis

Each signal below is reported in the output file for every matched row. This section explains what each signal means and why it has the value it does for this particular match.

euc_distance 0.0363

The best-match distance is **0.0363** — very small, indicating a close match. On its own this looks like a high-quality result. The problem is not the distance but the direction: the selected row is almost as close to Target B as it is to Target A.

per_feature_contribution (best match)

The Rank 1 supplemental row is extremely close to Target A on all features — the total distance is tiny and the per-feature contributions reflect small differences spread across dimensions. The contribution table alone does not signal a problem. The MNN check below is what reveals the issue.

Feature	Share of distance
Dragon Sightings	0.0%
Avg. Wizard Age	100.0%
Pct. Leprechaun Cottages	0.0%
Goblin Family Units	0.0%

nndr 0.0351

$\text{NNDR} = d_1/d_2 = 0.0363 \div 1.0324 = \mathbf{0.0351}$. This is well below the 0.80 threshold. The selected supplemental row is clearly the closest candidate — there is no ambiguity about which supplemental row wins. The MNN failure is a separate problem: the issue is not which supplemental row was chosen but whether this target should have claimed it.

near_miss_count 0

0 supplemental row(s) fall within the near-miss threshold. No near misses — the selected row is clearly the closest supplemental candidate. This reinforces that the MNN failure here is not about supplemental ambiguity but about target competition.

mnn_confirmed False

Confirmed: False. This is the key signal for this scenario. Running the match in reverse — from the selected supplemental row back across all target rows — the nearest target is **Target B**, not Target A. The supplemental row was assigned to Target A by the forward search, but it sits even closer to Target B. The asymmetry does not prove which assignment is correct — it flags that two targets are competing for the same supplemental row, so the researcher should review both assignments before use.

repeats 1

No exact tie — a single row sits alone at the minimum distance (the count includes the chosen match itself, so 1 means a unique winner).

smd (per feature) [0.0000, 0.0000, 0.0000, 0.0000]

SMD is computed across all matched target-supplemental pairs in a full run. With only one target row being explained here, it is reported as 0. In a production run, widespread MNN failures would appear as elevated SMD on the features driving the competition between target records.

flags MNN not confirmed - supplemental row is closer to a different target; this record may have no valid match

Flags: MNN not confirmed - supplemental row is closer to a different target; this record may have no valid match. The MNN flag is the primary concern. The researcher should inspect both Target A and Target B alongside the Rank 1 supplemental row and decide which assignment is correct for their analysis. In many cases the right action is to leave Target A unmatched rather than keep a pairing the reverse search does not support.

Data note: Values in this example are adapted from U.S. Census Bureau American Community Survey (ACS) public data. Column names have been replaced with illustrative labels for demonstration purposes. This document is intended for researcher education and internal system validation.